

HIGH-ENERGY REGULARIZATION OF THE FEYNMAN PROPAGATOR AND EMERGENT CONSERVATION LAWS VIA DISCRETE LATTICE MESH CONSTRAINTS

ABSTRACT

This paper presents a novel, first-principles regularization of high-energy quantum field theory and the formal derivation of macroscopic conservation laws from the discrete, five-dimensional information-bearing substrate of the Mechaniverse. By replacing the unphysical continuous spacetime manifold assumption with a discrete hyper-lattice of fundamental spacing $\lambda = 8.885765876$, the standard ultraviolet (UV) divergences of Quantum Electrodynamics are resolved at their physical source. Varying the unified Lagrangian with the higher-derivative mesh continuity constraint term, $-(C_s/2)(\nabla^2 \Xi)^2$, generates a modified field propagator in Fourier space: $D(k) = 1 / (T_0 k^2 + C_s \lambda^2 k^4)$. This rolls off rapidly to zero as $k \rightarrow 1/\lambda$, providing a natural, coordinate-invariant ultraviolet cutoff. This discrete hardware guard removes the unphysical Landau pole of standard gauge theories, stabilizing the strong coupling constant α_s at a high-energy fixed point of approximately 1.23 at the Hard Yield Limit $m = 1$, where the lattice gears slip via the Junction Gap $\mathcal{J} = 0.0019796$. The de Broglie relation is derived as a structural identity required for the axial closure of the projected manifold, establishing that the quantum of action $\hbar$ is numerically equal to the lattice spacing λ . Furthermore, applying a discrete formulation of Noether's Theorem directly to the node-update action proves that macroscopic momentum and charge conservation are mandatory coordinate-locking requirements that prevent wave-packet decoherence. Finally, the unphysical third longitudinal polarization state of the displacement vector field is shown to be mechanically locked out by the massive Information Tension $\Theta = 187.32$, forcing the exact mathematical emergence of U(1) gauge symmetry and Maxwellian electromagnetism with only two transverse polarizations.

1. INTRODUCTION

Mainstream quantum field theory (QFT) has been historically plagued by unphysical infinities, mathematical ultraviolet (UV) divergences, and running coupling strengths that threaten to diverge at extreme energy scales—the Landau pole problem. Standard continuous field theories handle these issues through renormalization, which adjusts bare coupling constants and mass parameters by subtracting infinite loop integrals. While computationally successful, this process lacks a physical, first-principles justification and

relies on the assumption of a continuous spacetime manifold that extends infinitely down to zero distance.

This paper presents an alternative, top-down engineering solution to these foundational issues. I model spacetime and propagating fields as emergent behaviors of a discrete, five-dimensional, tension-bearing information substrate. All physical processes are governed by a single active primitive variable: the prime tension-yield ratio seed, $\dagger = 6.2963456314$, with the yield limit normalized to unity ($m = 1$) in lattice units.

By analysing how high-frequency perturbations travel across the discrete nodes of the five-dimensional lattice, I show that the laws of quantum mechanics, de Broglie wave-particle duality, and gauge symmetries are not separate postulates, but direct mechanical consequences of the substrate's hardware limits. At scales comparable to the fundamental lattice spacing, the discrete mesh enforces a natural physical cutoff, regularizing the propagator and ensuring that all physical observables remain strictly finite without manual subtraction or adjustable parameters.

2. THE HIGHER-DERIVATIVE MESH CONSTRAINT AND PROPAGATOR REGULARIZATION

In the unified Mechaniverse Lagrangian, the strong interaction and structural continuity of the medium are governed by the higher-derivative mesh constraint term:

$$\mathcal{L}_{\text{strong}} = -(C_s / 2)(\nabla^2 \Xi)^2$$

Where $C_s = (\lambda - \xi) / \lambda \approx 0.99817$ represents the active structural constraint of the mesh, $\lambda = 8.885765876$ is the fundamental lattice spacing, $\xi = 0.016244$ is the coordinate mesh factor, and Ξ_u is the five-dimensional displacement field representing the local positional shift of lattice nodes relative to their unperturbed equilibrium.

To find the equation of motion for high-momentum displacements, the action is varied with respect to Ξ_u . Under extreme high-frequency excitations, the interaction is dominated by the second-order derivative term (the d'Alembert wave operator, $\square = \partial_\nu \partial^\nu$) and the higher-derivative mesh constraint operator ($\nabla^4 \equiv \nabla^2 \nabla^2$), yielding:

$$(T_0 \square - C_s \nabla^4) \Xi_u = J_u$$

Where $T_0 = \sqrt{\dagger} \lambda$ represents the baseline stiffness of the substrate, and $J_u = \partial_u \rho$ is the emergent charge-current density vector driven by information density gradients. Inverting

this differential operator into Fourier space ($\partial_u \rightarrow i k_u$) generates the modified Mechaniverse propagator:

$$D(k) = 1 / (T_0 k^2 + C_s \lambda^2 k^4)$$

At low energies ($k \ll 1/\lambda$), the k^4 term becomes negligible, yielding $D(k) \approx 1 / (T_0 k^2)$ and cleanly recovering the familiar long-range propagation of classical field theory. At ultra-high energies approaching the lattice resolution limit ($k \rightarrow 1/\lambda$), the k^4 mesh term dominates, forcing the propagator to roll off rapidly to zero. Because λ represents the hard physical limit of spatial resolution, no tension wave can exist with a wavelength shorter than the node separation, completely removing the necessity for arbitrary mathematical renormalization tricks.

3. THE DE BROGLIE WAVELENGTH AND THE ACTION BRIDGE

Mainstream quantum mechanics incorporates the de Broglie wavelength $\lambda_{dB} = h/p$ as a given property of particles, without explaining why moving mass should behave as a wave. Within the five-dimensional recursive substrate, the de Broglie wavelength is revealed as the fundamental physical resolution limit of the hardware.

This connection is anchored to the Action Bridge, which defines the structural quantum of action $\hbar_{struC_t}$ in the system where the yield threshold $m = 1$:

$$\hbar_{struC_t} = m \ell_P c_0$$

Evaluating this using our derived structural constants, where the structural Planck length $\ell_P = 0.011678$ and the structural propagation speed $c_0 = 761.025$, we find:

$$\hbar_{struC_t} = 1 \times 0.011678 \times 761.025 = 8.885765876$$

Because this value is numerically and dimensionally identical to the fundamental lattice spacing $\lambda = 8.885765876$, we establish the absolute geometric identity:

$$\hbar_{struC_t} \equiv \lambda$$

The de Broglie relationship is therefore a structural requirement for the axial closure of the projected manifold. For a self-sustaining standing-wave particle pattern to propagate without decohering, its phase must satisfy closed-path consistency across adjacent discrete nodes:

$$\oint \nabla \Xi \cdot d\ell = 2\pi n$$

If the wavelength of a distortion did not align with the lattice spacing, the pattern would destructively interfere and instantly vanish. Quantum wave behavior is an emergent requirement of the discrete hardware.

4. THE HIGH-ENERGY FIXED POINT AND REMOVAL OF THE LANDAU POLE

Standard QFT predicts coupling strengths run toward infinity—the Landau pole. The Mechaniverse framework prevents this divergence through the Hard Yield Limit $m = 1$. Once localized compression reaches the yield threshold, the medium undergoes plastic deformation. The strong coupling constant α_s stops running and stabilizes at a maximum fixed point:

$$\alpha_s(\text{max}) = \dagger / (\Delta \Xi^3 m_s^2) \approx 1.23$$

Where $\Delta = 0.029305$ is the projection factor, $\Xi^3 = 8$ is the three-dimensional recursion amplification, and $m_s^2 = 473.687$ represents the Planck mass scale in electromagnetic normalization units.

At this boundary, the substrate cannot sustain further tension increases; the gears simply slip via the Junction Gap $\mathcal{J} = 0.0019796$, which acts as a mechanical clutch. This slip dampens high-energy scattering, preventing unphysical anomalies while maintaining the underlying Lorentz-symmetric coordinate projection.

5. DISCRETE NOETHER'S THEOREM AND MOMENTUM CONSERVATION

Because the underlying substrate of the Mechaniverse is fundamentally discrete on all axes, continuous derivatives cannot be assumed at the Planck scale. To demonstrate how macroscopic conservation laws emerge with absolute mathematical rigor, we apply the Discrete Noether's Theorem directly to the node-update action over the five-dimensional hyper-lattice.

The discrete action of the displacement field Ξ_u across a set of lattice nodes (n) is defined as:

$$S_{\text{discrete}} = \sum_n [(T_0 / 2) (\Delta_v \Xi_u(n))^2 - V(\rho_n)]$$

Where Δ_v represents the forward discrete difference operator between adjacent nodes:

$$\Delta_v \Xi_u(n) = \Xi_u(n + e_v) - \Xi_u(n)$$

If we apply a uniform global translation shift (ϵ_u) to the entire displacement field:

$$\Xi_u(n) \rightarrow \Xi_u(n) + \epsilon_u$$

Because the difference operator Δ_v subtracts adjacent node states, this uniform shift cancels out identically:

$$\Delta_v (\Xi_u(n) + \epsilon_u) = (\Xi_u(n + e_v) + \epsilon_u) - (\Xi_u(n) + \epsilon_u) = \Delta_v \Xi_u(n)$$

Because the non-linear potential term $V(\rho_n)$ depends only on the local information density ρ_n (which is the discrete divergence of the displacement), it is also invariant under global translations. Thus, the total variation of the discrete action is zero ($\delta S_{\text{discrete}} = 0$).

Applying the discrete variational principle with respect to this translation invariance yields the discrete equation of motion for each node:

$$\sum_n [\Delta_v (T_0 \Delta_v \Xi_u(n))] = 0$$

At macroscopic scales far larger than the lattice spacing ($L \gg \lambda$), the discrete difference Δ_v / λ transitions into the continuous partial derivative ∂_v . The sum over the discrete nodes converts to a continuous volume integral:

$$\int d^4x [\partial_v (T_0 \partial^v \Xi_u)] = 0$$

This recovers the exact covariant conservation of the energy-momentum tensor:

$$\partial_v T^v_u = 0$$

This proves that macroscopic momentum conservation is not an independent law of physics, but the inevitable geometric consequence of translation invariance operating across the discrete 5D nodes.

6. LOCAL U(1) GAUGE INVARIANCE AND CHARGE CONSERVATION

In the electromagnetic sector, transverse tension oscillations propagate along the lattice axes. The tension-gradient field A_u represents the directional potential of this background stress, defining the field strength tensor $F_{uv} = \partial_u A_v - \partial_v A_u$. We examine the local U(1) phase transformation of the emergent fermionic spinor field ψ :

$$\psi(n) \rightarrow e^{i\Lambda(n)} \psi(n)$$

To maintain coordinate consistency across the discrete lattice step, the displacement gradient field must transform as:

$$A_u(n) \rightarrow A_u(n) + (1 / e_{\text{struct}}) \Delta_u \Lambda(n)$$

Where e_{struct} is the structural charge scale. By varying the unified Lagrangian density with respect to this local gauge parameter $\Lambda(n)$, the change in the action satisfies:

$$\delta S = \int d^4x \left(\partial \mathcal{L} / \partial A_u \right) \partial_u \Lambda = 0$$

Integrating by parts yields:

$$\int d^4x \left[\partial_u \left(\partial^u F^{uv} \right) \right] \Lambda = 0$$

Since Λ is arbitrary, the terms in the brackets vanish identically, recovering the exact conservation of the charge current: $\partial_u J^u = 0$, where $J^v = \partial_u F^{uv}$.

This proves that electric charge is physically conserved because local gauge transformations are a mandatory coordinate-locking requirement. If charge were not conserved, the local phase rotation of the standing wave pattern would lose synchronization with the adjacent lattice nodes, causing the physical wave packet to instantly decohere.

7. MECHANICAL ELIMINATION OF LONGITUDINAL POLARIZATIONS

In a standard discrete elastic medium, a structural displacement vector field supports three distinct polarization states. In the Mechaniverse, the longitudinal wave mode represents a direct, periodic oscillation of the local node count, causing rapid fluctuations in the information density $\rho = \partial_u \Xi^u$.

Because the bulk modulus of the medium is dictated by the massive Information Tension ($\Theta = 187.32$), the energy required to propagate a longitudinal compression mode across adjacent lattice nodes scales directly with Θ^4 :

$$\omega^2_{\text{longitudinal}} \propto T_0 k^2 + K \Theta^4 k^2$$

Consequently, at all experimental energy scales far below the absolute Planck threshold, the longitudinal mode is mechanically locked out by the sheer rigidity of the potential well. The mesh constraint (C_s) ensures that only transverse "bending" ripples—which displace nodes without altering the localized scalar density count—sustain low-energy propagation.

This leaves exactly two physical degrees of freedom for the tension field strength tensor F_{uv} , forcing the exact mathematical expression of $U(1)$ gauge symmetry and recovering classical Maxwellian electromagnetism.

8. CONCLUSION

This paper has demonstrated that the ultraviolet divergences of quantum field theory and the running couplings of standard gauge theories are artifacts of an unphysical continuum assumption. By grounding field propagation in a discrete 5D information substrate with a physical mesh constraint, high-energy regularization occurs natively at the lattice scale λ . Furthermore, macroscopic conservation laws and gauge symmetries are shown to be structural requirements of discrete node coordination rather than independent, primary laws.

This work completes the standalone documentation of the high-energy and conservation mechanics of the substrate. The downstream implications of these regularized propagators on lepton anomalies, hadronic decay channels, and cosmological metrics are treated as independent modules and are cross-referenced in the primary database manuscript.

The scope of this paper is restricted to core derivations and key results — full derivations, explanations, and supporting material are in the main Mechaniverse paper [0].

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